

Rowan Barbalich

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Education

Stevens Institute of Technology

Master of Mechanical Engineering with Control Systems Concentration

5/20/2026

Related Coursework: Autonomous Navigation for Mobile Robots, Mobile Micro robotic Systems, Modern Control

Engineering, Analytical Dynamics, Engineering Analysis, Engineering Modeling and Simulation

Bachelor of Mechanical Engineering, Minor of Green Engineering

5/19/2025

Related Coursework: Statics and Introduction to Engineering Mechanics, Circuits and Systems,

Thermodynamics, Design of Dynamical Systems, Thermal Engineering, Manufacturing Processes and Systems, Field Sustainable Systems with Sensors, Engineering Economics and Project Management, Fluid Mechanics

Technical Skills

CAD/Simulation: PID Tuning, Closed-Loop Feedback, Sensor Integration, Embedded systems

CAD/Simulation: SolidWorks, Blender, MATLAB, ANSYS (Workbench, APDL, Fluent), AutoCAD

Fabrication: CNC Milling, Lathe Operation, TIG Welding, Autodesk Fusion

Programming: MATLAB, C++, Python, Arduino

Other Tools: MS Office Suite, Google Suite, Adobe Photoshop/Illustrator, AI (Claude, ChatGPT, etc.)

Academic Projects

Programmable Gas Chromatography Oven

- Led a 5-person mechanical design team through three project phases, running weekly progress meetings and enforcing scheduling accountability to deliver a fully functional prototype achieving temperature ramp rates up to 400°C in under twenty minutes.
- Modeled the full oven assembly in SolidWorks and validated the design through SolidWorks Simulation studies — structural analysis under 2 atm internal pressure, thermal distribution, and airflow — to confirm structural integrity and safe operating temperatures before fabrication.
- Fabricated inner and outer steel enclosures via CNC milling and lathe operation; completed final assembly using riveting and TIG welding.

Autonomous Campus Tour Robot

- Designed and fabricated an autonomous mobile robot in SolidWorks and manufactured the chassis via 3D printing to house LiDAR, ultrasonic sensors, and control electronics.
- Integrated YDLIDAR G4 laser scanner for odometry, along with multiple ultrasonic sensors for real-time obstacle detection and collision avoidance in a physical obstacle course.
- Developed a path-planning algorithm in Arduino using live x-y coordinate data streamed for active navigation.

PD/PID Compensator Design for Closed Loop Pitch Control

- Analyzed a closed-loop pitch-control architecture for an unmanned submersible using root locus and Bode plot analysis to evaluate stability margins across varying gain values.
- Designed a PD compensator via damping ratio and natural frequency calculations to meet a $\leq 15\%$ overshoot, ≤ 15 -second settling time specification.
- Extended the design to a PID compensator to eliminate steady-state error, iteratively tuning gain in MATLAB Control System Designer to achieve 13.95% overshoot and a 14.5-second settling time.

Autonomous Navigation: RRT Path Planning Under Localization Uncertainty

- Implemented a Rapidly exploring Random Tree (RRT) path planner in MATLAB with Kalman filter state propagation to quantify localization uncertainty along candidate paths through an obstacle field.
- Evaluated the tradeoff between obstacle proximity (for localization accuracy) and collision risk from obstacle-inflation buffers to inform uncertainty-aware path-planning decisions.

Experience

MHE Engineering, D.P.C.

June 2023 – August 2023, Milford PA

- Produced engineering and site plans in AutoCAD for municipal and private infrastructure projects including roadway drainage and structural renovation.
- Conducted field inspections to verify construction compliance with engineering specifications.
- Analyzed and organized watermain material data in Excel for the Town of Port Jervis, NY, supporting infrastructure condition assessment.

Other Experience: SAE Baja, Club Lacrosse Vice President, Engineers without Borders